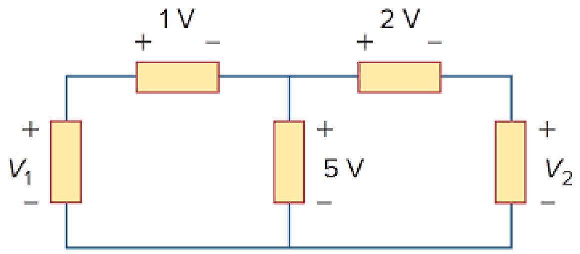
Chapter 2, Problem 11P

Problem

In the circuit of Fig. 2.75, calculate *V*1 and *V*2.

Figure 2.75:



Step-by-step solution

Step 1 of 4

Consider the following circuit diagram:

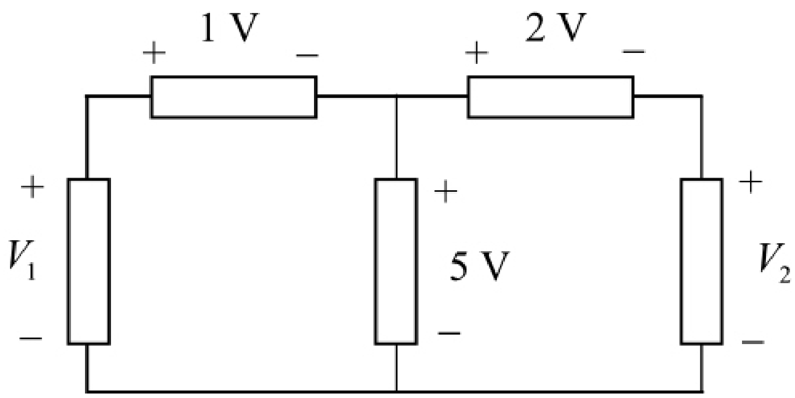


Figure 1

Step 2 of 4

Redraw the circuit with two loops.

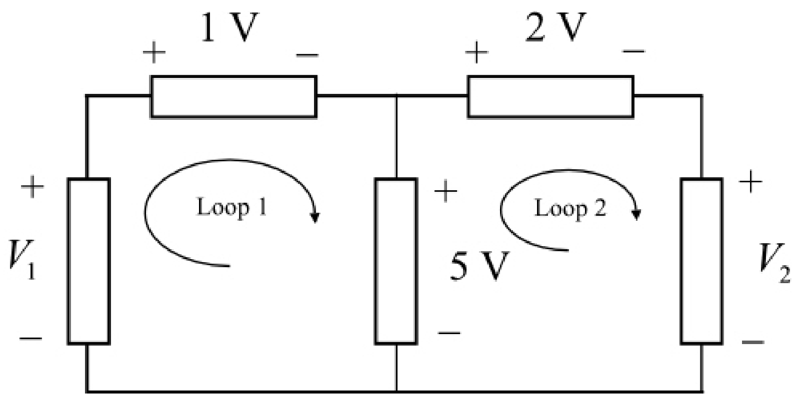


Figure 2

Step 3 of 4

Apply Kirchhoff's voltage law around the loop 1.

$$-V_1 + 1 + 5 = 0$$
$$V_1 = 6 \text{ V}$$

Therefore the voltage V_1 is $\boxed{6 \text{ V}}$.

Step 4 of 4

Apply Kirchhoff's voltage law around the loop 2.

$$-5 + 2 + V_2 = 0$$
$$V_2 = 3 \text{ V}$$

Therefore the voltage V_2 is $\boxed{3 \text{ V}}$.